

PROBABILITY-EXPLANATORY NOTES

Solutions for questions 1 to 20:

1. We know that when n coins are tossed, the probability of getting exactly 'r' heads is ${}^nC_r/2^n$.

(i) $\therefore$ The probability of getting four heads, when six coins are tossed is $\frac{{}^6C_4}{2^6} = \frac{15}{64}$. Ans: $\left(\frac{15}{64}\right)$

(ii) The probability of getting at least two heads
 $= 1 - \text{probability of getting less than two heads.}$
 $= 1 - [p(0 \text{ head}) + p(1 \text{ head})]$
 $= 1 - \left(\frac{{}^6C_0}{2^6} + \frac{{}^6C_1}{2^6}\right) = 1 - \frac{1+6}{64} = \frac{57}{64}$.
 Ans: $\left(\frac{57}{64}\right)$

2. When a dice is rolled two times total outcomes are $6 \times 6 = 36$.

(i) Favourable outcomes that the sum is 9 = {(3, 6), (4, 5), (5, 4), (6, 3)} = i.e. 4.
 Required probability = $\frac{4}{36} = \frac{1}{9}$. Ans: $\left(\frac{1}{9}\right)$

(ii) Probability that the sum is at least 4 = $1 - \text{probability that the sum is less than 4.}$
 $= 1 - \{p(\text{sum } 2) + p(\text{sum } 3)\}$
 $= 1 - \left(\frac{1}{36} + \frac{2}{36}\right) = \frac{11}{12}$. Ans: $\left(\frac{11}{12}\right)$

3. (i) Probability of drawing a spade card is $\frac{{}^{13}C_1}{{}^{52}C_1} = \frac{1}{4}$.
 Ans: $\left(\frac{1}{4}\right)$

(ii) Probability of drawing a numbered card is $\frac{{}^{36}C_1}{{}^{52}C_1} = \frac{9}{13}$.
 Ans: $\left(\frac{9}{13}\right)$

(iii) Probability of drawing a honour card is $\frac{{}^{16}C_1}{{}^{52}C_1} = \frac{4}{13}$.
 Ans: $\left(\frac{4}{13}\right)$

(iv) Let $S \cup H$ be the event that the card is a spade or a honour.
 $P(S \cup H) = P(S) + P(H) - P(S \cap H)$
 $= \frac{13}{52} + \frac{16}{52} - \frac{4}{52} = \frac{25}{52}$. Ans: $\left(\frac{25}{52}\right)$

4. (i) Two cards can be selected from 52 cards in ${}^{52}C_2$ ways.

Number of favourable cases that the two cards drawn are from the same suit = ${}^4C_1 \times {}^{13}C_2 = 4 \times {}^{13}C_2$
 $\therefore$ Required probability = $\frac{4 \times {}^{13}C_2}{{}^{52}C_2} = \frac{4}{17}$.

Ans: $\left(\frac{4}{17}\right)$

- (ii) Favourable cases that the two cards drawn are from different suits = ${}^4C_2 \times {}^{13}C_1 \cdot {}^{13}C_1$.

Required probability = $\frac{{}^4C_2 \times {}^{13}C_1 \cdot {}^{13}C_1}{{}^{52}C_2} = \frac{13}{17}$.

Ans: $\left(\frac{13}{17}\right)$

5. Two cards can be selected from 36 numbered cards in ${}^{36}C_2$ ways. The favourable cases that the two cards are black and different number is ${}^2C_1 \times {}^8C_1 \times {}^9C_1$

$\therefore$ Favourable cases

$= {}^2C_1 \times {}^8C_1 \times {}^9C_1 + {}^2C_1 \times {}^8C_1 \times {}^9C_1 = 16 \times 18$

Required probability = $\frac{288}{{}^{36}C_2} = \frac{16}{35}$. Ans: $\left(\frac{16}{35}\right)$

6. (i) Total outcomes = ${}^{15}C_3$
 Favourable cases = ${}^6C_1 \times {}^5C_1 \times {}^4C_1$
 Required probability = $\frac{6 \times 5 \times 4}{{}^{15}C_3} = \frac{24}{91}$. Ans: $\left(\frac{24}{91}\right)$

- (ii) Three balls can be selected from fifteen balls one after another without replacement in $15 \times 14 \times 13$ ways. The favourable cases that the three balls are of different colours is $6 \times 5 \times 4$.

The three balls can be in any order.

Total favourable cases = $3! \times 6 \times 5 \times 4$

Required probability = $\frac{3! \times 6 \times 5 \times 4}{15 \times 14 \times 13} = \frac{24}{91}$.

Ans: $\left(\frac{24}{91}\right)$

- (iii) Three balls can be selected from fifteen balls one after another with replacement in $15 \times 15 \times 15$ ways. The favourable cases that they are different colour is $3! \times 6 \times 5 \times 4$

$\therefore$ required probability = $\frac{3! \times 6 \times 5 \times 4}{15 \times 15 \times 15} = \frac{16}{75}$.

Ans: $\left(\frac{16}{75}\right)$

7. (i) The probability that bag x is selected is $\frac{1}{2}$.
 Two balls can be selected from bag x in 9C_2 ways. Favourable cases for the two balls being of the same colour is ${}^5C_2 + {}^4C_2$

The probability = $\frac{{}^5C_2 + {}^4C_2}{{}^9C_2}$

The probability that the bag x is selected and two balls of the same colour can be drawn from it is $\frac{1}{2} \left(\frac{{}^5C_2 + {}^4C_2}{{}^9C_2} \right)$

Similarly, two balls of same colour can be selected from bag y in $\frac{1}{2} \frac{{}^5C_2 + {}^4C_2}{{}^9C_2}$

$\therefore$ Required probability = $\frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{16}{36} = \frac{4}{9}$.

Ans: $\left(\frac{4}{9}\right)$

- (ii) The total outcomes that one ball from each bag can be selected is ${}^9C_1 \times {}^9C_1$ ways = 81 ways

Favourable cases are: white ball from bag x and blue ball from bag y or blue ball from bag x and white ball from bag y.

$\Rightarrow$ It can be done in ${}^5C_1 {}^5C_1 + {}^4C_1 {}^4C_1 = (25 + 16)$ ways

Required probability = $\frac{25+16}{81} = \frac{41}{81}$. Ans: $\left(\frac{41}{81}\right)$

8. Let $A \cup B$ be the event that the number is divisible by 4 or 5.
 $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{25}{100} + \frac{20}{100} - \frac{5}{100}$
 $= \frac{40}{100} = \frac{2}{5}$.
 Ans: $\left(\frac{2}{5}\right)$

9. Number of defective nuts made by factory I is 5% of 2000 i.e. = 100.
 Number of defective nuts made by factory II is 15% of 5000 i.e. = 750.

Required probability = $\frac{750}{100 + 750} = \frac{750}{850} = \frac{15}{17}$.

Ans: $\left(\frac{15}{17}\right)$

10. Let A be the event that A completes the task, B be the event that B completes the task and, C be the event that C completes the task.

Given $P(A) : P(\bar{A}) = 3 : 4 \Rightarrow P(A) = \frac{3}{7}$; $P(\bar{A}) = \frac{4}{7}$

$P(\bar{B}) : P(B) = 2 : 5 \Rightarrow P(B) = \frac{5}{7}$; $P(\bar{B}) = \frac{2}{7}$.

$P(C) : P(\bar{C}) = 2 : 5 \Rightarrow P(C) = \frac{2}{7}$; $P(\bar{C}) = \frac{5}{7}$

(i) $P(\text{All of them complete the task})$
 $= P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

$= \frac{3}{7} \times \frac{5}{7} \times \frac{2}{7} = \frac{30}{343}$.
 Ans: $\left(\frac{30}{343}\right)$

(ii) $P(\text{None of them complete the task})$
 $= P(\bar{A} \cap \bar{B} \cap \bar{C})$

$= P(\bar{A}) P(\bar{B}) P(\bar{C}) = \frac{4}{7} \times \frac{2}{7} \times \frac{5}{7} = \frac{40}{343}$.

Ans: $\left(\frac{40}{343}\right)$

(iii) Probability of at least one of them complete the task
 $= 1 - \text{probability that none of them complete the task}$
 $= 1 - \frac{40}{343} = \frac{303}{343}$.
 Ans: $\left(\frac{303}{343}\right)$

(iv) Exactly one of them complete the task is denoted by
 $A \cap \bar{B} \cap \bar{C}$ or $\bar{A} \cap B \cap \bar{C}$ or $\bar{A} \cap \bar{B} \cap C$.
 Required Probability =
 $P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)$
 $= P(A) P(\bar{B}) P(\bar{C}) + P(\bar{A}) P(B) P(\bar{C}) + P(\bar{A}) P(\bar{B}) P(C)$
 $= \frac{3}{7} \times \frac{2}{7} \times \frac{5}{7} + \frac{4}{7} \times \frac{5}{7} \times \frac{5}{7} + \frac{4}{7} \times \frac{2}{7} \times \frac{2}{7}$
 $= \frac{30 + 100 + 16}{343} = \frac{146}{343}$.
 Ans: $\left(\frac{146}{343}\right)$

11. $P(A) = \frac{3}{5}$.
 $P(\bar{A}) = \frac{2}{5}$.
 $P(B) = \frac{4}{7}$.
 $P(\bar{B}) = \frac{3}{7}$.

If at least one of them solves the problem, then the problem will be solved. It is denoted by $A \cup B$.
 Required probability = $1 - \text{probability that none of them solve the problem}$.

$P(A \cup B) = 1 - P(\bar{A}) P(\bar{B}) = 1 - \frac{2}{5} \times \frac{3}{7} = \frac{29}{35}$.

Ans: $\left(\frac{29}{35}\right)$

12. Each person can be born on any one of the 7 days.
 $\therefore$ Total outcomes = $7 \times 7 \times 7 \times 7 \times 7 \times 7$.
 Out of six persons any two persons can be born on same day. This is possible in ${}^6C_2 \times 7$ ways.
 Remaining four persons can be born on different days. This can happen in $6 \times 5 \times 4 \times 3$ ways.
 $\therefore$ Required probability

$= \frac{{}^6C_2 \times 7 \times 6 \times 5 \times 4 \times 3}{7 \times 7 \times 7 \times 7 \times 7 \times 7} = \frac{15}{2} \times \left(\frac{6!}{7^5}\right)$.
 Ans: $\left(\frac{15}{2} \times \left(\frac{6!}{7^5}\right)\right)$

13. Let C be the event that the question can be answered correctly. $P(C) = \frac{2}{5}$; $P(\bar{C}) = \frac{3}{5}$.

Let A be the event that atleast one question can solved correctly. $\bar{A}$ be the event that no question is answered correctly. $P(\bar{A}) = \left(\frac{3}{5}\right)^8$

$P(A) = 1 - P(\bar{A}) = 1 - \left(\frac{3}{5}\right)^8$.
 Ans: $\left(1 - \left(\frac{3}{5}\right)^8\right)$

14. Let p be the probability that the dice shows composite number. $p = \frac{2}{6} = \frac{1}{3}$

Let q be the event that the dice shows a non-composite number

$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$.

B can win the game in the following cases.

A lose B win or A lose B lose A lose B wins or A lose B lose A lose B lose A lose B win or

and so on

This is denoted by

The probability that B wins the game

$= qp + q^3p + q^5p + q^7p + \dots$

$= qp(1 + q^2 + q^4 + q^6 + \dots) = qp \frac{1}{1 - q^2} = \frac{\frac{2}{3} \times \frac{1}{3}}{1 - \left(\frac{2}{3}\right)^2} = \frac{2}{5}$.

Ans: $\left(\frac{2}{5}\right)$

15. (i) The probability that the number is a multiple of 3 is $\frac{6}{20}$ i.e. $\frac{3}{10}$.

The probability that the number is a multiple of 7 is $\frac{2}{20}$ i.e. $\frac{1}{10}$.

Expected value = $\frac{3}{10} \times 15 + \frac{1}{10} \times 20 = \frac{65}{10} = ₹6.50$

Average profit = Expected value - entry fee = ₹6.50 - 4 = ₹2.50.
 Ans: (2.50)

(ii) Expected amount = $\frac{1}{2} \times 15 - \frac{1}{2} \times 25 = -5$

As expected value is -5, he has to pay ₹5.

Ans: (5)

16. (i) Let p be the probability that the man can take a step forward. $p = \frac{3}{7}$. Let q be the probability that the man can take a step backward $q = \frac{4}{7}$.

The probability that he is one step away from starting point = 5 steps forward 4 steps backward or 4 steps forward 5 steps backward. = ${}^9C_5 p^5 \cdot q^4 + {}^9C_4 p^4 q^5$
 $= {}^9C_4 p^4 q^4 (p + q)$

$= 126 \left(\frac{3}{7}\right)^4 \left(\frac{4}{7}\right)^4 (1) = (126) \frac{(12)^4}{7^8} = \frac{3^6 \cdot 2^9}{7^7}$.

Ans: $\left(\frac{3^6 \cdot 2^9}{7^7}\right)$

(ii) In 8 steps he will never be one step away from starting point

$\therefore$ Required probability is 0.

Ans: (0)

17. Total outcomes = ${}^{36}C_4$
 Favourable cases = 9

Required probability = $\frac{9}{{}^{36}C_4}$.

Ans: $\left(\frac{9}{{}^{36}C_4}\right)$

18. Two squares can be selected from 64 squares in ${}^{64}C_2$ ways.
Favourable cases are
 $4({}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 + {}^7C_2) + 2 \cdot {}^8C_2$
 $\therefore$ Required probability
$$= \frac{4({}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 + {}^7C_2) + 2 \cdot {}^8C_2}{{}^{64}C_2} = \frac{280 \times 2 \times 1}{64 \times 63}$$
$$= \frac{5}{36}.$$
 Ans: $\left(\frac{5}{36}\right)$

19. Let red ball be transferred into bag B. It's probability is $\frac{6}{13}$.
Now in bag B there are 6 red balls and 9 black balls.
Probability of drawing a black ball from bag B = $\frac{9}{15}$.
Hence the probability = $\frac{6}{13} \times \frac{9}{15}$.
Let black ball be transferred into bag B. It's probability is $\frac{7}{13}$
Now in bag B, there are 5 red balls 10 black balls.

Probability of drawing a black ball and from bag B = $\frac{10}{15}$
Hence the probability = $\frac{7}{13} \times \frac{10}{15}$.
Required probability = $\frac{6}{13} \times \frac{9}{15} + \frac{7}{13} \times \frac{10}{15} = \frac{124}{195}$.
Ans: $\left(\frac{124}{195}\right)$

20. Probability of attempting at least one Maths question
= 1 - probability of attempting no Maths question is
attempted = $1 - \frac{{}^5C_4}{{}^{11}C_4}$
$$= 1 - \frac{5}{330} = \frac{65}{66}.$$
 Ans: $\left(\frac{65}{66}\right)$

ADDITIONAL SOLUTIONS FOR PRACTICE

Solutions for questions 1 to 5:

1. The mayor may be from city A or from city B.
The probability that he is from city A and is an officer is
 $\frac{3}{5} \times \frac{7}{10}$.
The probability that the mayor is from city B and is an officer
is $\frac{2}{5} \times \frac{4}{13}$.
 $\therefore$ Required probability
$$= \frac{3}{5} \times \frac{7}{10} + \frac{2}{5} \times \frac{4}{13} = \frac{273 + 80}{130 \times 5} = \frac{353}{650}.$$
 Ans: $\left(\frac{353}{650}\right)$

2. Probability of bomb hitting the target is $\frac{3}{4}$.
Probability of bomb not hitting the target is $\frac{1}{4}$.
Let there be n bombs dropped.
The probability that at least one bomb hits the target
$$= 1 - \text{probability that no bomb hits the target} = 1 - \left(\frac{1}{4}\right)^n.$$

Given $1 - \left(\frac{1}{4}\right)^n \geq 0.95$
$$\left(\frac{1}{4}\right)^n \leq 0.05$$

 $\therefore$ The minimum value of n is 3. Ans: (3)

3. 'Let q be probability of tails appearing on any of the coins.
(q = 1 - p)
Given ${}^{50}C_{25} p^{25} q^{25} = {}^{50}C_{26} p^{26} q^{24}$
 ${}^{50}C_{25} q = {}^{50}C_{26} p$
$$\frac{50!}{25!25!} q = \frac{50!}{24!26!} p \Rightarrow \frac{26}{25} q = p$$

$$\Rightarrow \frac{26}{25} (1 - p) = p \Rightarrow 26(1 - p) = 25p$$

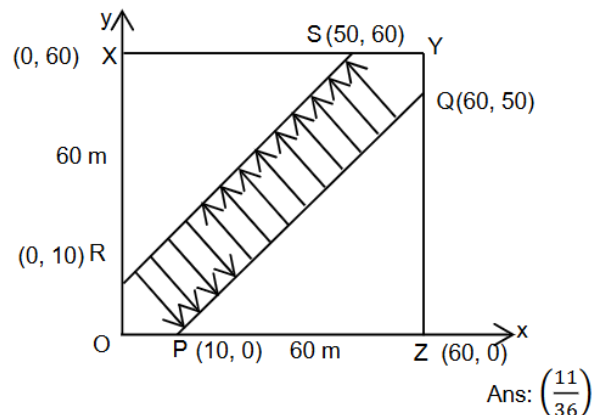
$$\Rightarrow 51p = 26 \Rightarrow p = \frac{26}{51}.$$
 Ans: $\left(\frac{26}{51}\right)$

4. For every five throws, the dice shows four times even and one time odd
It means out of 2000 throws it shows an even number in 1600 throws and shows an odd number in 400 throws
 $1600(4) - 400(2) = ₹5600$
Amount Rakesh would realize in throwing 2000 times
Since he has to pay ₹2500, it will be profit. Rakesh will play the game.
Ans: (Play the game)

5. OXYZ represents time space in which A and B have to meet.
Now $|A - B| \leq 10$ as the time gap between A and B is maximum 10 minutes.
 $A - B \leq 10$ or $B - A \leq 10$
The shaded region represents the favourable area for the two friends to meet

The required probability = $\frac{\text{Area of the shaded region}}{\text{Area of the region OXYZ}}$

$$= \frac{\text{Areas OXYZ} - (\text{area of } \triangle PZQ + \text{area of } \triangle RXS)}{\text{Area of the region OXYZ}}$$
$$= \frac{60 \times 60 - 2\left(\frac{1}{2} \times 50 \times 50\right)}{60 \times 60} = \frac{11}{36}.$$



Ans: $\left(\frac{11}{36}\right)$